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Image-Informed Urban Propagation and Route-Level Body Exposure at 15 GHz

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ABSTRACT Street-level radiofrequency exposure at 15 GHz depends on the route geometry, surface materials, transmitter locations, and body orientation. This study maps 360-degree street images onto a photogrammetric city mesh and assigns surface materials around five selected pedestrian routes. A common source model sets the expected number of transmitters from their areal density and distributes them along the visible roofline in proportion to its physical length. All exposure values are normalized per unit areal density times effective isotropic radiated power. The calculation treats direct paths and one specular reflection exactly. It estimates one diffuse reflection and then stops the path. The directional fields are coupled to a 56,024-element Duke body surface. The five routes contain 73 route points. Each point uses 16 independent replicas, 200,000 primary rays per replica, and 4,096 fixed first-diffuse angular cells. In a controlled open-square case, the maximum bounced-term difference between the adjoint estimator and deterministic quadrature is 0.0616 dB, and the maximum total-transport difference between the adjoint and forward tracers is 0.0344 dB. The route medians of normalized whole-body specific absorption rate differ by a factor of 13.34. Direct transport is the largest term at all 67 nonshadowed points. First-diffuse transport gives the only nonzero modeled contribution at six shadowed points. The maximum 12-to-16-replica change in total transfer is 0.0436 dB across the five sites, although lower-tail estimates in Mexico City and Tokyo are less stable. The result applies to five selected routes under the fixed model and does not estimate city-wide or deployed-network exposure.

INDEX TERMS Body dosimetry, image-informed propagation, ray tracing, street imagery, urban propagation, whole-body specific absorption rate.

I. INTRODUCTION

URBAN wireless systems operate in a built environment that strongly shapes radio propagation [1]. Propagation can change over a few meters along a pedestrian route. A person can move from direct visibility of a roofline to a region where buildings block the direct field and reflected power arrives from another direction. The surface materials then affect how much power is returned to the street [1], [2]. This local variation also matters after propagation. Whole-body absorption depends on the arrival direction and on the orientation of the body, so one incident-power value at one receiver position does not describe exposure along a route. A route calculation must retain position and arrival direction until the field is coupled to the body [3].

Urban radio ray tracing provides detailed paths between specified transmitters and receivers [4]. It has been used for city-scale downlink exposure with base-station locations from public deployment records [5], and in hybrid propagation and anatomical dosimetry calculations along an outdoor user path [6]. Stochastic site models provide a different description when individual transmitters are not fixed [7], while normalized body coefficients separate incident fields from later exposure calculations [8]. Together, these studies include deterministic urban transport, spatial source models, and body coupling. Because these studies use different source assumptions, their reported quantities are not directly comparable without a common source law.

Street images can supply surface information that is absent from an untextured city mesh. The Vistas dataset provides a street-scene taxonomy for dense semantic segmentation [9]. Kamari *et al.* segment street-level images, project the resulting material classes onto city geometry, and use that geometry in millimeter-wave ray tracing [10]. Xia *et al.* use semantic point-cloud classification and detailed scene reconstruction for outdoor urban ray tracing at 2.8 GHz [11]. Image-informed city modeling is therefore established. The present work does not claim semantic segmentation, material classification, or image-to-geometry projection as new. It uses these operations to form a traceable material map around fixed pedestrian routes. The map keeps the geometry-based material wherever the images give no reliable label.

The calculation combines these established parts in one fixed model. First, each 360-degree street image is aligned with the same city mesh used for ray tracing. The image labels are then projected onto that mesh to make a material map. Second, the same transmitter model is used at every site, with the number of transmitters set independently of how finely the roofline is divided. Third, direct, specular, and diffuse power stays separate until its arrival direction is coupled to the body. The transport calculation treats direct paths and one specular reflection exactly, estimates one diffuse reflection, and then stops. The route points are fixed case studies rather than a population sample, and the roofline transmitters are a model rather than a measured deployment. To the best of the authors' knowledge, prior work has not combined aligned 360-degree street images, a common roofline transmitter model, these separate transport components, and directional body coupling along fixed routes.

The study applies this method at 15 GHz to five selected routes with 73 fixed route points. Each result is normalized per unit active-source areal density and per unit equivalent isotropically radiated power. The calculation treats direct paths and one specular reflection exactly, then estimates one diffuse reflection by next-event estimation. It omits further reflections. The reported route distributions are therefore results for five fixed routes under this source and transport model. They do not estimate population exposure or deployed-network exposure, and they do not rank the five cities.

For the first time, one fixed-route calculation combines the following three contributions.

1) Aligned 360-degree street images supply traceable material labels to the same city mesh used for the propagation calculation. Surfaces without a reliable image label keep their geometry-based material.
2) A normalized roofline transmitter model and transport calculation preserve direct, one-reflection specular, and first-diffuse power and direction until whole-body coupling.
3) A five-site application reports fixed-route exposure distributions, controlled first-diffuse validation, component closure, replica convergence, and the retained transport at the six route points with no direct or one-reflection specular contribution.

II. CONFIGURATION AND SURFACE MAPPING

The study configuration is shown in Fig. 1. A 360-degree street image is aligned with a photogrammetric city mesh cropped to a 250 m radius. Object and material labels from the image are projected onto the visible mesh triangles without changing their geometry. The roofline visible from the route gives the possible transmitter locations. Fixed points along the pedestrian route give the receiver locations, and the anatomical phantom faces along the direction of travel at each point.

TABLE 1. Five fixed routes and prepared-scene numerical campaign times

Site	Route points	Route span (m)	Wall time (s)
Korenmarkt	10	49.04	29.79
Prague	22	119.39	69.02
Madrid	14	73.47	40.70
Mexico City	11	60.79	30.92
Tokyo Hachiko	16	87.38	50.11
Total	73	390.07	220.54

Table 1 lists the five selected routes. The point counts and spans come from the verified route files used for the five-site data set. Each route follows a connected corridor covered by aligned 360-degree street images. The calculation samples fixed points along that corridor, including interpolated points between image locations. The image count and route-point count can therefore differ. The 73 route points are fixed observations rather than a random sample of pedestrians or places. The phantom faces along the walk, so a different route would change both its position and orientation.

The two image models have separate roles. Mask2Former assigns a Vistas object class to every image pixel [9], [12]. These classes include buildings, roads, people, vehicles, and vegetation. SAM 3 then tests relevant material and vegetation labels inside compatible object regions [13]. The known camera position and viewing direction project both sets of labels onto the visible city mesh. Repeated observations are combined into one material map, and every mapped triangle stays linked to its original image. A mesh triangle changes material only when the object and material labels agree and pass the acceptance tests. All other triangles keep their geometry-based material. The prompts, image-alignment tests, rejected labels, and mapping rules are given in the supplementary material.

Fig. 2 shows the computation and the files checked before it runs. A hash-verified manifest lists the aligned images, material map, city mesh, route, roofline, body, sampler, and transport settings. Only files that match this list enter the five-site data set. The transport calculation keeps direct, first-order specular, and first-diffuse contributions separate until body coupling. The five manifests contain 210 verified entries. Across the resulting 1,168 directional body fields, the largest additive-closure residual is $1.735 \times 10^{-18}\,\mathrm{m}^{-2}$.

III. FIXED-ROUTE EXPOSURE METHOD

The calculation uses a fixed route, city mesh, material map, and roofline source curve. Every route point uses 15 GHz and a crop with radius $R_{\mathrm{crop}} = 250$ m, which gives $A_{\mathrm{crop}} = \pi R_{\mathrm{crop}}^2 = 196{,}349.54\,\mathrm{m}^2$. The receiver position $\mathbf{x}$ is also the reciprocal ray origin and body reference point. The phantom faces along the local direction of travel. Index i denotes a roofline segment, $r_i(\mathbf{x})$ is its range to the receiver, and $\mathbf{r}$ is the position of a body surface element. These assumptions stay fixed across replicas and sites.

The exact positions of future transmitters are unknown, but elevated rooflines are plausible street-facing locations. The model therefore distributes the expected transmitters uni-

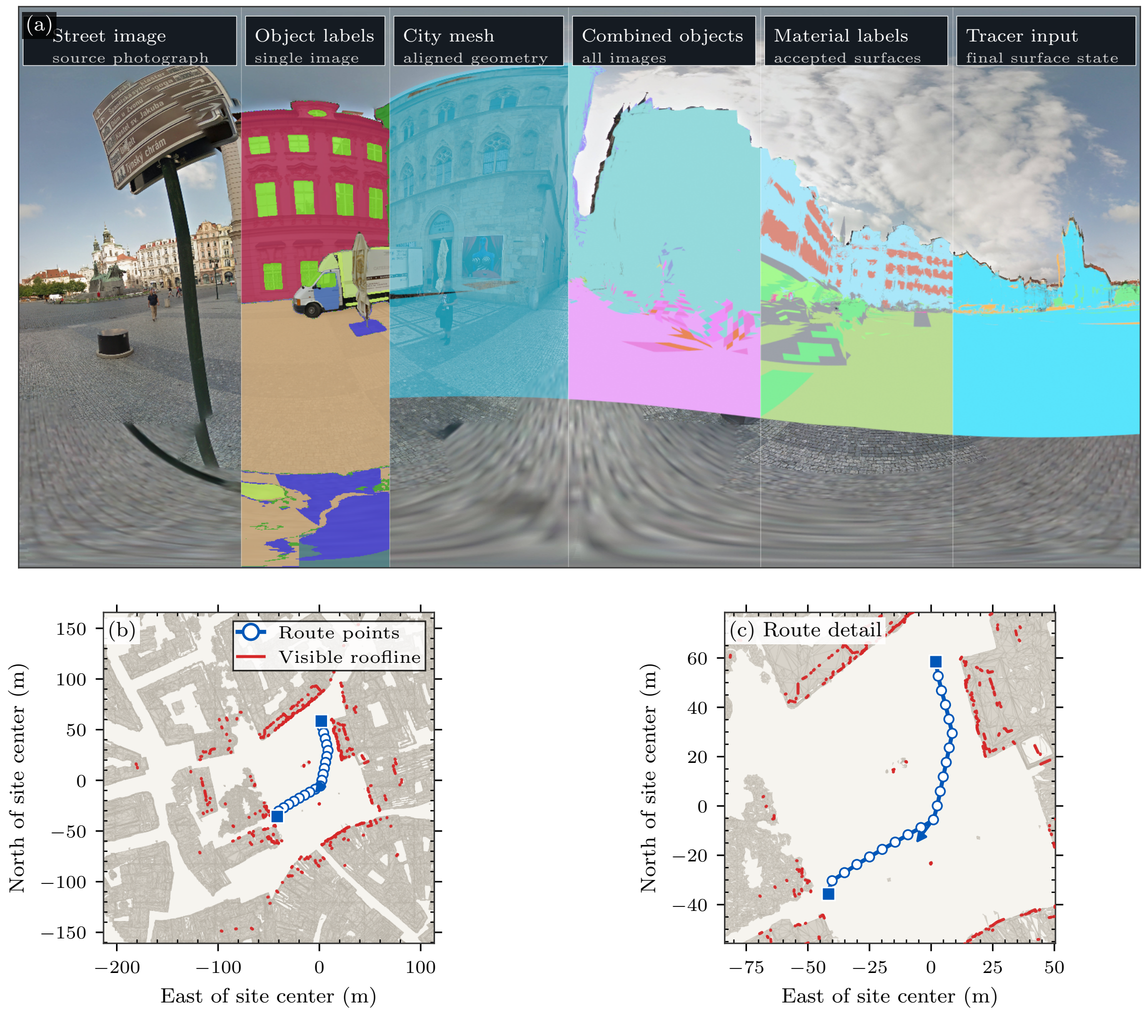

FIGURE 1. **Study configuration at Prague Old Town Square. Panel (a) follows a 360-degree street image through object segmentation, projection onto the city mesh, and conversion to the material map used by the tracer. Panel (b) shows the 22 route points and the visible roofline on a plan view of the city. Panel (c) enlarges the route. The arrow gives the phantom's direction along the walk.**

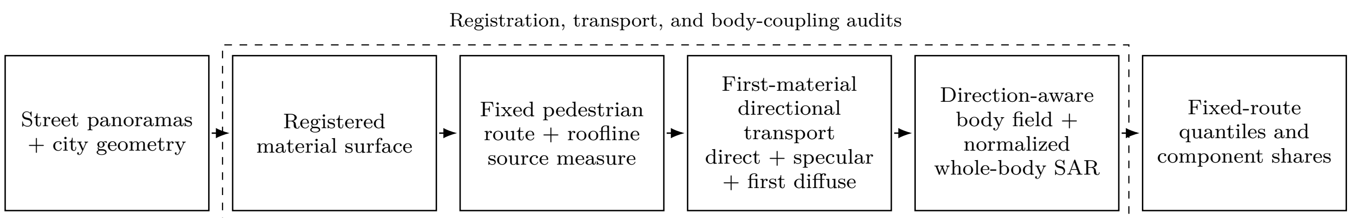

FIGURE 2. **Flowchart of the computation and input checks. Aligned 360-degree street images and the city mesh give the material map. The fixed route and visible roofline give receiver and possible transmitter positions. Transport retains direct, first-order specular, and first-diffuse terms. A hash-verified file list fixes every input to the five-site result.**

formly per unit physical length of the roofline visible from the route. Let ρ_A be the expected active-site density and let A_{crop} be the crop area. For a roofline segment with endpoints $\mathbf{p}_i$ and $\mathbf{p}_{i+1}$, ℓ_i is its physical three-dimensional length. The expected number of transmitters and the fraction assigned to segment i are

$$N_{\text{site}} = \rho_A A_{\text{crop}}, \qquad p_i = \frac{\ell_i}{\sum_j \ell_j}, \qquad \ell_i = \|\mathbf{p}_{i+1} - \mathbf{p}_i\|_2 \,. \tag{1}$$

Thus, N_{site} sets the number of transmitters and p_i assigns a fraction of that number to segment i. Dividing by the total roofline length ensures that splitting one segment into smaller numerical pieces does not add transmitters. The baseline uses three-dimensional length rather than horizontal projected length.

An unobstructed reference keeps network scale separate from scene visibility. The reference includes the inverse-square geometry of the complete source curve:

$$D_{\mathrm{ref}}(\mathbf{x}) = \sum_i \frac{p_i}{r_i(\mathbf{x})^2}, \qquad S_{\mathrm{ref}}(\mathbf{x}) = \frac{A_{\mathrm{crop}} D_{\mathrm{ref}}(\mathbf{x})}{4\pi}. \tag{2}$$

No visibility test enters D_{ref}. The reported transfer and body endpoints are normalized per unit $\rho_A P_{\mathrm{EIRP}}$. Multiplication by $\rho_A P_{\mathrm{EIRP}}$ gives a physical scale only for a deployment that follows the same roofline transmitter model. The calculation omits transmitters and interactions outside the crop.

The directional transfer has direct, one-reflection specular, and one-reflection diffuse parts. Let $\mathcal{D}$ contain the visible direct paths, and let $\mathcal{S}_1$ contain the accepted one-reflection specular paths. Their normalized powers are $\alpha_a = m_a^{(\mathrm{d})}/D_{\mathrm{ref}}$ and $\beta_b = m_b^{(\mathrm{s})}/D_{\mathrm{ref}}$. The diffuse estimate uses normalized angular-cell powers $\widehat{\gamma}_q = \widehat{m}_q^{(\mathrm{f})}/D_{\mathrm{ref}}$. With $\delta_{\widehat{\mathbf{k}}}$ denoting a unit point mass in physical arrival direction $\widehat{\mathbf{k}}$, the directional distribution is

$$\begin{aligned} \mu_{\mathbf{x}} = &\sum_{a\in\mathcal{D}} \alpha_a \delta_{\widehat{\mathbf{k}}_a^{(\mathrm{d})}} \\ &+ \sum_{b\in\mathcal{S}_1} \beta_b \delta_{\widehat{\mathbf{k}}_b^{(\mathrm{s})}} \\ &+ \sum_{q=1}^{Q} \widehat{\gamma}_q \delta_{\widehat{\mathbf{k}}_q^{(\mathrm{f})}}, \qquad Q = 4096. \end{aligned} \tag{3}$$

The first two sums retain the exact directions and powers of the direct and specular paths. They are not projected onto the angular grid. Only diffuse power is accumulated in the Q Fibonacci cells. Rays start at the receiver and travel outward to the first blocking surface, as shown in Fig. 3. Next-event estimation then tests a connection from that surface to every roofline segment [14]. The material model combines unpolarized Fresnel power with a Rayleigh roughness term. The remaining power enters a Lambertian diffuse term, and the components add incoherently. The sampled path ends after this diffuse reflection, and the calculation omits further specular reflections. Woody vegetation identified in the material map does not block rays because the city mesh has no canopy volume. The supplementary material gives the full laws and parameter values.

Direction remains explicit until body coupling. For outward body-surface normal $\widehat{\mathbf{n}}(\mathbf{r})$, define $g(\mathbf{r},\widehat{\mathbf{k}}) = [\widehat{\mathbf{n}}(\mathbf{r}) \cdot (-\widehat{\mathbf{k}})]_+$. The normalized absorbed power density is

$$\begin{aligned} \widetilde{S}_{\mathrm{ab}}(\mathbf{r},\mathbf{x}) &\equiv \frac{S_{\mathrm{ab}}(\mathbf{r},\mathbf{x})}{\rho_A P_{\mathrm{EIRP}}} \\ &= T_0 S_{\mathrm{ref}}(\mathbf{x}) \Bigg[\sum_{a\in\mathcal{D}} \alpha_a g(\mathbf{r},\widehat{\mathbf{k}}_a^{(\mathrm{d})}) \\ &\qquad + \sum_{b\in\mathcal{S}_1} \beta_b g(\mathbf{r},\widehat{\mathbf{k}}_b^{(\mathrm{s})}) \\ &\qquad + \sum_{q=1}^{Q} \widehat{\gamma}_q g(\mathbf{r},\widehat{\mathbf{k}}_q^{(\mathrm{f})}) \Bigg]. \end{aligned} \tag{4}$$

Here, T_0 is the normal-incidence power-transmission coefficient obtained from the IT'IS tissue parameters at 15 GHz [15]. The implementation applies this one-sided local-incidence coupling to the Duke anatomical mesh with the published level-2 dosimetry kernel [16], [17]. Thus, $\widetilde{S}_{\mathrm{ab}}$ is dimensionless. For the area $A_{\mathbf{r}}$ at position $\mathbf{r}$ and body mass m_{body}, the normalized integrated quantities are

$$\begin{aligned} \widetilde{P}_{\mathrm{abs}}(\mathbf{x}) &\equiv \frac{P_{\mathrm{abs}}(\mathbf{x})}{\rho_A P_{\mathrm{EIRP}}} \\ &= \sum_{\mathbf{r}} A_{\mathbf{r}} \widetilde{S}_{\mathrm{ab}}(\mathbf{r},\mathbf{x}), \\ \widetilde{\mathrm{SAR}}_{\mathrm{wb}}(\mathbf{x}) &\equiv \frac{\mathrm{SAR}_{\mathrm{wb}}(\mathbf{x})}{\rho_A P_{\mathrm{EIRP}}} \\ &= \frac{\widetilde{P}_{\mathrm{abs}}(\mathbf{x})}{m_{\mathrm{body}}}. \end{aligned} \tag{5}$$

$\widetilde{P}_{\mathrm{abs}}$ has units m^2, and $\widetilde{\mathrm{SAR}}_{\mathrm{wb}}$ has units $\mathrm{m}^2\ \mathrm{kg}^{-1}$.

Each replica launches 200,000 independent and identically distributed primary rays per route point and accumulates first-diffuse power in 4,096 fixed Fibonacci output cells. Exact direct and specular paths bypass this grid. The output cells do not control the launch directions. The calculations use 16 replicas with seeds 7 through 22 and retain cumulative results after 4, 8, 12, and 16 replicas. Only the first-diffuse estimate varies between replicas. Scalar quantities and body fields are averaged before route statistics are computed. Route quantiles use NumPy linear interpolation over equally weighted route points. The plotted empirical distributions use positions $(\mathrm{rank} - 0.5)/n$. These quantities describe a selected route and do not estimate a pedestrian population. Multipath surplus is computed only where direct transfer is positive. A zero-direct route point remains in the whole-body SAR distribution but has no finite surplus value.

IV. VALIDATION AND RESULTS

Table 1 defines the five fixed routes and their 73 calculation points. Every site uses 15 GHz, a photogrammetric city mesh cropped to a 250 m radius, and the Duke body model with 56,024 surface elements and a 72.4 kg mass. The phantom faces along the walk. Each point has 16 independent replicas with seeds 7 through 22, and each replica uses 200,000 IID primary rays and 4,096 fixed output directions for the first-diffuse term. The resulting 1,168 fields contain 233.6 million

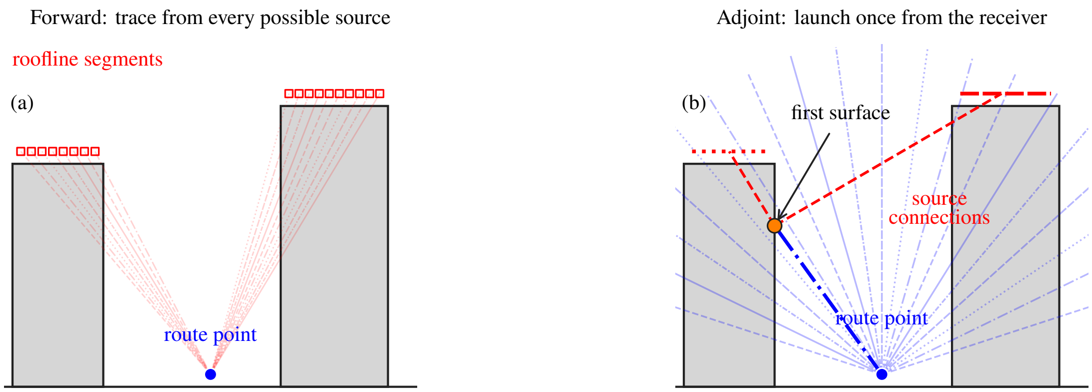

FIGURE 3. **Forward and adjoint sampling for the first-diffuse term. (a) A forward calculation launches rays from every roofline segment toward the route point. (b) The adjoint calculation launches rays once from the route point. At the first blocking surface, next-event estimation tests connections to the roofline.**

primary rays. Calculation time for a prepared site ranges from 29.79 to 69.02 s on one A6000 GPU. These times exclude image acquisition, image-to-mesh alignment, depth estimation, and material mapping because the full preparation time was not recorded.

The calculation manifests list 42 files per site, and all 210 file hashes pass verification. The direct, exact order-1 specular, and first-diffuse fields sum to the stored total with a maximum absolute residual of 1.735×10^{-18} m^{-2} across all 1,168 fields. The GPU body-coupling result also agrees with the double-precision CPU reference to a maximum relative difference of 6.64×10^{-16} in the verified benchmark. These checks confirm the input files, addition of components, and agreement between CPU and GPU calculations. They do not externally validate the complete city model.

The controlled comparison in Fig. 4 tests the first-diffuse estimator before the city results. The open-square scene has 27 sources, six receivers, eight triangles, and one diffuse reflection. It has no specular reflection, refraction, or diffraction. Deterministic surface quadrature uses 2,097,152 samples. The adjoint estimate uses 50,000 primary rays for each of four seeds. An independent Sionna RT forward calculation uses 50,000 samples per source for each of three seeds. The maximum difference between the adjoint estimate and quadrature is 0.0616 dB for the reflected term. A separate test of total transport gives a maximum adjoint-to-Sionna difference of 0.0344 dB. Fig. 4 plots the one-reflection transfer and its error relative to quadrature, not the total-transport test. The comparison checks first-diffuse normalization, visibility, inverse-square loss, and cosine terms in this one-reflection scene. It does not cover the image-derived material map, exact specular transport, or their combination in a city.

Fig. 5 shows the fixed-route empirical distributions and route-mean component shares. Table 2 gives the central summaries and finite multipath surplus. Normalized whole-body SAR is reported in $\mathrm{m}^2\ \mathrm{kg}^{-1}$ per unit $\rho_A P_{\mathrm{EIRP}}$. A physical deployment value therefore requires multiplication by its areal source density and EIRP. Panel (a) includes all 73 route points and marks the six points with zero direct and zero order-1 specular transfer. The largest route median is 13.34 times the smallest. The selected routes do not define a city ranking. Mexico City and Tokyo Hachiko have the widest ranges along a route and the lowest tails.

The component shares in Fig. 5(b) are additive shares of route-mean whole-body SAR. Direct transport is the largest body contribution at all 67 nonshadowed points. Exact order-1 specular transport is never the largest. Mexico City points 0, 1, and 3 and Tokyo Hachiko points 13, 14, and 15 have zero direct and zero order-1 specular transport. First-diffuse transport is the only nonzero modeled contribution at these six points. Across all 73 points, the pooled component medians are 77.662% direct, 21.391% exact order-1 specular, and 0.419% first diffuse. They do not sum to 100% because each component has a separate median over the 73 route points. The small first-diffuse median therefore does not describe the six shadowed points.

The path audit assigns each retained material interaction to an image-mapped or geometry-based material. Pooled over the five routes and 16 seeds, image-mapped surfaces account for 75.903% of the non-direct body-coupled whole-body SAR contribution. The corresponding shares are 80.643% for exact order-1 specular transport and 13.337% for first-diffuse transport. The supplementary material gives the complete split by material source.

The nested 12-to-16-replica comparison separates the central route statistic from the lower tail. Every route-median whole-body SAR changes by at most 5.90×10^{-5} dB. The largest lower-decile point changes are 0.032226 dB in Mexico City and 0.017636 dB in Tokyo Hachiko. The maximum pointwise total-transfer changes in Table 2 reach 0.043625 and 0.019732 dB at these two sites. At 16 replicas, their 90th-percentile total-transfer standard errors are 0.1461 and 0.0310 dB, respectively. Central route statistics are stable under the retained estimator.

A separate calculation retains the exact verified 16-replica prefix and continues every route through 64 replicas. Be-

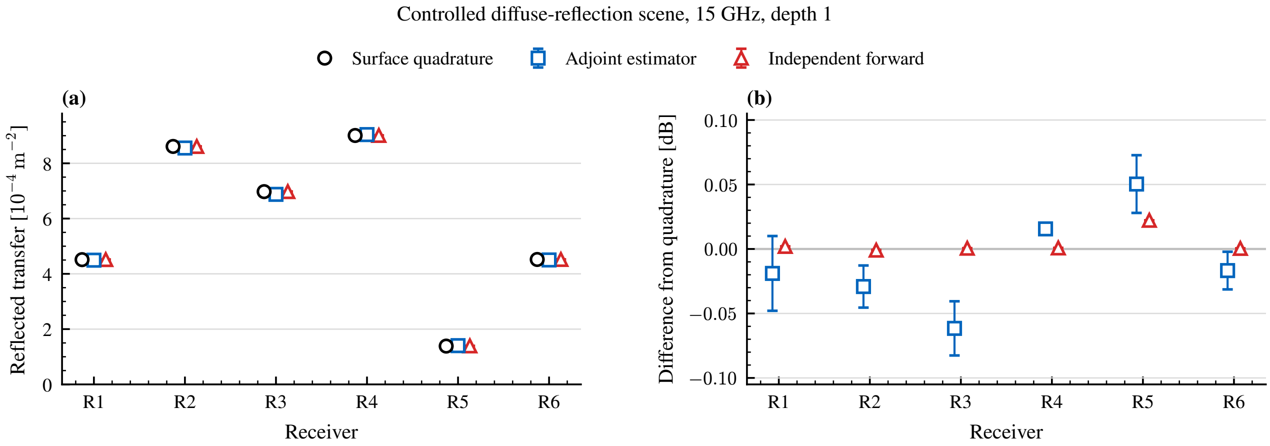

FIGURE 4. **Controlled validation of the first-diffuse transfer. (a) Deterministic surface quadrature, the adjoint estimate, and the independent Sionna RT forward calculation at six receivers. (b) Signed error relative to quadrature. Error bars give the standard error across four adjoint seeds and three Sionna seeds.**

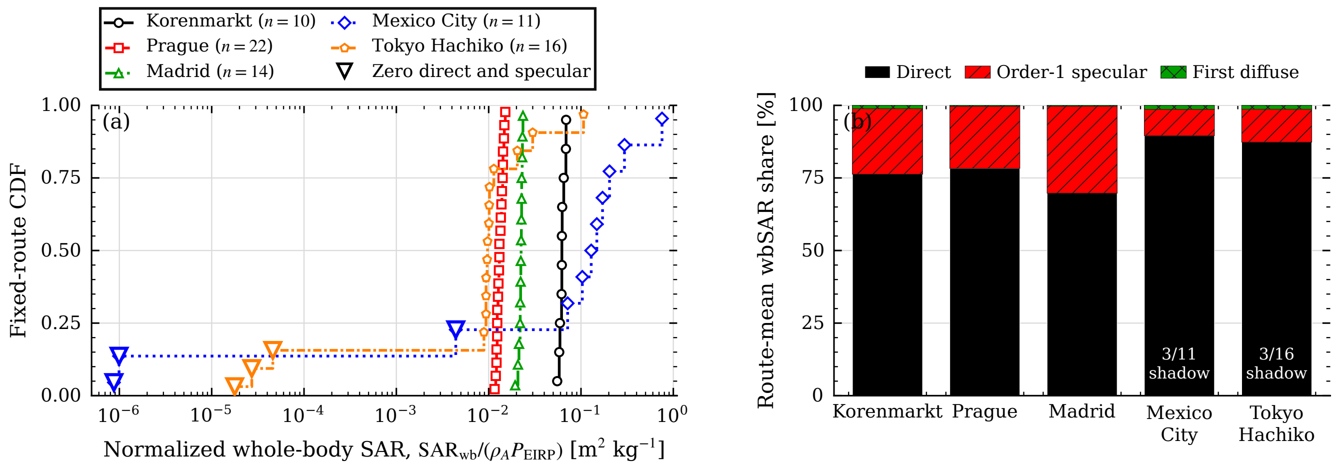

FIGURE 5. **Normalized whole-body SAR on the five fixed routes. (a) Midpoint empirical CDFs include all 73 route points. Hollow triangles mark the six points with zero direct and order-1 specular transfer. (b) Additive contributions to route-mean whole-body SAR. The routes are fixed case studies rather than city or population samples.**

TABLE 2. **Fixed-route exposure summary. Whole-body SAR quantiles are normalized per unit $\rho_A P_{\mathrm{EIRP}}$ and have units m^2 kg^{-1}. Surplus uses finite-direct points only. The last column is the maximum pointwise total-transfer change from 12 to 16 replicas.**

Site	q_{10}	q_{50}	q_{90}	Median surplus [dB]	Zero direct and order-1 specular	Max total-transfer change [dB]
Korenmarkt	0.057833	0.062045	0.068672	1.108	0	0.0000819
Prague	0.012157	0.013073	0.014594	1.061	0	0.0001559
Madrid	0.020913	0.022381	0.023171	1.534	0	0.0004083
Mexico City	9.92×10^{-7}	0.129062	0.295798	0.721	3	0.043625
Tokyo Hachiko	3.66×10^{-5}	0.009674	0.025200	0.828	3	0.019732

tween 48 and 64 replicas, the whole-body SAR q_{10} changes by 0.00344 dB in Mexico City and 0.00491 dB in Tokyo Hachiko. The largest change among their six shadowed route points is 0.0125 and 0.0104 dB, respectively. Both lower tails satisfy the stated aggregate stability criteria through 64 replicas. Mexico City still shows rare-event first-diffuse behavior, with a maximum-to-median positive replica contribution ratio of 5738. This result remains specific to the fixed routes and excludes route-selection and city-sampling uncertainty. The complete nested comparison is provided in the supplementary material.

A paired material-map control was completed for Madrid and Mexico City. The control replaces all image-mapped materials with the geometry-based materials while retaining the mesh, route, roofline transmitter model, body, seeds, sampling budget, and transport steps. Each reported change is $10\log_{10}(x_{\mathrm{image}}/x_{\mathrm{geometry}})$. The image-to-geometry changes in normalized whole-body SAR at Madrid are $+0.233$, $+0.249$, and $+0.269$ dB for q_{10}, q_{50}, and q_{90}. The corresponding changes at Mexico City are $+24.84$, -0.158, and $+0.104$ dB. The large Mexico City q_{10} change comes from its three shadowed points, where both totals are near zero and first diffuse is the only nonzero modeled contribution. The direct term is identical in every pair. At Madrid, the image-derived map changes the route-median specular component by $+1.89$ dB and the first-diffuse component by -12.43 dB, although the total median changes by $+0.249$ dB. The control tests the complete material map, including its material parameters and the treatment of woody vegetation as nonblocking. It does not measure material accuracy or isolate reflectance alone. The supplementary material gives pointwise and component-level comparisons.

V. DISCUSSION

The factor of 13.34 between the largest and smallest route medians shows that body exposure differs among the selected routes after the same source-density and EIRP normalization. This contrast includes the geometry, visible roofline, mapped materials, and visibility conditions of each selected route. The empirical distribution for each site therefore describes only that fixed route and its calculation points. It does not estimate a city distribution or define a ranking of the five cities. The analysis reports changes along each pedestrian route while keeping the statistical unit clear.

The three modeled components explain the local changes within the route distributions. A direct path is present at 67 route points. The exact order-1 specular term adds one surface reflection. The first-diffuse term connects the receiver to a roofline transmitter through one blocking surface. After body coupling, the first-diffuse contribution is small at most nonshadowed route points. The six remaining points have zero direct and zero order-1 specular contributions. The first-diffuse contribution is the only nonzero modeled contribution at those points. Therefore, its small contribution at most nonshadowed positions does not make it dispensable at shadowed positions.

The paired material control tests the effect of image-derived materials on the body results. The direct term is identical between the image-derived and geometry-based cases, so the observed changes arise from the materials used by the reflected and diffuse terms. In Madrid, the specular and first-diffuse component changes have opposite signs, while the route-median normalized whole-body SAR changes by 0.249 dB. The Mexico City route median changes by -0.158 dB. Its much larger lower-tail ratio comes from three shadowed route points where both estimates are close to zero, and it is not a stable central effect. The paired cases differ in their assigned materials and in their treatment of woody canopy. The image-derived case treats identified woody canopy as pass-through without attenuation because the city mesh has no canopy volume. The comparison therefore measures sensitivity to both choices and does not establish the accuracy of the assigned materials.

Several limits restrict the interpretation. The study covers five selected routes at 15 GHz and one body model that faces along each walk. The reported scale is per unit areal source density and EIRP. Scaling to a specific deployment is valid only if its transmitter positions follow the assumed roofline model. The transport model contains exact direct transport, exact order-1 specular transport, and one first-diffuse event. It stops at that diffuse event and omits all later interactions as well as higher-order specular paths. The controlled comparison validates the first-diffuse component in a depth-1 case. It does not validate the city calculations that also contain image-derived materials and specular transport. The 16 replicas quantify estimator randomness. They do not include uncertainty in image-to-mesh alignment, geometry, material labels, route choice, transmitter placement, body shape, or body orientation. The repeated five-site computation takes less than 70 s per prepared site on the tested GPU, but image acquisition and material mapping take longer and do not yet have one complete timing record. A separate geometric fixed-grid diagnostic in the supplementary material provides a broader configuration check and remains separate from these fixed-route results.

The first priority is source calibration and validation of the city calculation. Calibration should use a measured site source distribution. Outdoor field measurements should then test the directional field before body coupling. Second, higher specular orders and paths after a diffuse event can be added through paired studies that report their change in whole-body SAR, variance, and computation time. Third, several routes at the same site can quantify route-selection variation before the site set is enlarged. Other frequencies, body models, and orientations can then test the remaining range of validity. Each extension should also report how much of the surface reached by the modeled paths has an image-derived material.

VI. CONCLUSION

This study aligned 360-degree street images with a photogrammetric city mesh to map surface materials around fixed pedestrian routes. A common roofline transmitter model and direction-aware body coupling then gave normalized whole-body SAR along each route. Each result was normalized by $\rho_A P_{\mathrm{EIRP}}$. The transport model retained exact direct paths, exact order-1 specular paths, and the first diffuse reflection. In a controlled depth-1 case, the adjoint first-diffuse estimate had maximum bounced-transport errors of 0.0616 dB against deterministic surface quadrature and 0.0621 dB against Sionna RT forward tracing. The five scenes contained 73 route points. Their route-median values differed by a factor of 13.34. Direct transport was largest at 67 nonshadowed points, while first-diffuse transport was the only nonzero modeled contribution at all six shadowed points.

These results describe five selected routes at 15 GHz per unit active-source density and EIRP. They do not rank cities or predict deployed-network exposure. The comparison uses the fixed transmitter model, material map, Duke body facing along the walk, and one-diffuse-reflection limit. The 16-replica checks showed stable central route statistics, although the shadowed lower tails remained less stable. The distributions have no population weighting across each urban area and have greater estimator uncertainty at locally shadowed positions.

DATA AND CODE AVAILABILITY

The verified five-site data, manifests, analysis scripts, and figure scripts are retained with the study repository. A stable public archive with a versioned digital object identifier will be deposited before publication. The 360-degree street images and commercial photogrammetric tiles are governed by their providers' terms and are not redistributed. Their identifiers and the transformation manifests are retained to support reacquisition where the licenses permit it.

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OpenAI Codex [18] was used to draft and edit text in all manuscript sections and to assist with plotting code and manuscript checks. The author reviewed the generated text and code and verified the scientific claims, numerical values, references, and final text.

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